

Controller bench sheet

Flash the advanced build, prove the controller, then measure the LED string. Gate 1 of docs/10-commissioning.md.

Before you start

No cells connected for any of this. Use a current-limited bench supply, 200 mA limit, for VCC.
Step 2 needs the regulator and power LED removed (Gate 0) and the bench supply as the only power source.
Do not connect the panel or the charger. R3 sits at the board end, not out at the string.
Duty does not limit current: only R3 and the pin do. Verify at 4.20 V and 100% duty.

1 Flash the advanced build

Fuses are set once per board (internal 8 MHz, BOD 2.7 V, no bootloader). Routine flashing never writes them.

Disconnect cells, panel, UART and LED string. Power from one verified target-compatible ISP source only.

Action	Command / pass result
Open firmware folder	cd firmware/controller
Check the target	avrdude -c usbasp -p m328p -v -> signature 1E 95 0F 1E 95 16 = ATmega328PB: use -p m328pb, never -F
Compile advanced build	pio run -e advanced -> SUCCESS
Upload and verify	standalone current avrdude; never pio -t upload
First power-up check	USB-UART at 3.3 V into serial-header VCC -> banner; vdd within 0.05 V of meter

Disconnect all ISP wires first. Then remove the regulator and power LED (guide 04 step 5) and repeat the check.
After that, USB-UART is monitoring-only: target TX -> adapter RX, GND -> GND; adapter VCC, TX, DTR, CTS open.

Tool setup, fuses, Arduino-as-ISP fallback and troubleshooting: docs/08-firmware.md

2 Check the controller on a current-limited bench supply

Read the diagnostics through Tier 1 telemetry or a USB-UART adapter (RX and GND only) at 9600 baud.

Check	Expect
Startup after reset	Banner with version and build date, field guide, then diagnostics
VCC at 3.80 V	vdd within 0.05 V of the meter (config.h BANDGAP_V = 1.10 x Vmeter / Vreported)
D9 PWM frequency	1.9 to 2.0 kHz, confirming the 8 MHz clock
Pull D2 to GND	cause=BUTTON_TEST, output=100%, button timer counts down
Cover / uncover LDR	DUSK / DAWN confirmation reaches 300 s before mode changes

3 LED string current with R3 = 47 ohm

Meter in series with the string positive, lights at 100% duty (press D2 or cover the LDR). Expect = model:

Supply	Expect	Measured	R3 temp	Notes
3.40 V	11 mA	_____	_____	_____
3.70 V	15 mA	_____	_____	_____
4.20 V	21 mA	_____	_____	_____

Decision rule

Over 22 mA at 4.20 V: fit 56 ohm and repeat the three readings (expect about 18 / 13 / 9 mA).
Too bright or want longer run time: lower NIGHT_DUTY_PCT in config.h. Too dim: this design is at its limit; a brighter string needs a driver stage (docs/12-going-further.md).

Also confirm: R3 and the D9 pin stay at ambient; the string is dark with M2 unplugged and during reset.

For deployment, build pio run -e production (or standalone) and flash its .hex the same way.